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Program PANCO Version 3 - User's Manual

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Resumen/Summary

This Technical Report describes the utilisation procedure of the PANCO Program, Version 3.3.

The PANCO Program calculates reserve factors and applied and allowable stresses of uniform and non uniform metallic columns, with different support conditions under the action of axial and/or transversal loads.

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1 TECHNICAL SHEET OF THE PROGRAM

Technical Report: **ME-T-NT-110086**PROGRAMREFERENCEDATE

PANCO

March 2015

OBJECTIVE

This program was developed to columns, with different support conditions under the action of axial and/or transversal loads.

LANGUAGE

Fortran 77

COMPUTER

Sun Solaris

NUMBER OF LINES (APROX.)

6500 (Fortran, analysis core)

+ 4000 (Visual Basic.Net)

Total 10500

EXECUTION MODE

Console (input data in text file)

Interactive GUI

Notes:

- Blank lines and commented lines excluded
- Count for Visual Basic code excludes lines created by Visual Studio in designer.vb files

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VERSION

1.0

3.0

3.3

ADDITIONAL DOCUMENTS

Technical Report:

Nota Técnica NT – 00 – ECE - 042

Manual Teórico del Programa PANCO, Versión 1.0

Technical Report:

Nota Técnica NT – 00 – ECE - 043

Manual del Programador del Programa PANCO, Versión 1.0

2 ABBREVIATIONS AND SYMBOLS

Symbol	Definition
Et	Tension Elastic Modulus
Ec	Compression Elastic Modulus
G	Shear Elastic Modulus
Ftu	Ultimate allowable stress
Fty	Tension Yield allowable stress
Fcy	Compression Yield allowable stress
e	Ultimate Strain (in percentage)
L	Column length
A	Cross Section Area
I _{yy}	Cross Section Moment of Inertia about Y-axis
I _{zz}	Cross Section Moment of Inertia about Z-axis
J	Cross Section Torsion constant
R	Cross Section warping constant
y ₀	Y coordinate of the shear center respect to the center of gravity
z ₀	Z coordinate of the shear center respect to the center of gravity
F _{cc}	Crippling allowable stress
D _s	Distance from de Neutral Axis to the upper fiber (in the same plane that the transversal applied loads)
D _i	Distance from de Neutral Axis to the lower fiber (in the same plane that the transversal applied loads)
N _g	Number of sections for geometry analysis along the column
x _i	X Coordinate of the section <i>i</i>
A _i	Cross section area at Section <i>i</i>
I _i	Moment of Inertia about the axis perpendicular to the applied bending loads plane at Section <i>i</i>
D _{S_i}	Distance from the Neutral Axis to the upper fiber (in the same plane that the transversal applied loads) at Section <i>i</i>
D _{I_i}	Distance from the Neutral Axis to the lower fiber (in the same plane that the transversal applied loads) at Section <i>i</i>
WP	Working plane
N _{Px}	Number of Sections where a discrete axial load is applied
x _{Pxi}	X Coordinate where the discrete axial load <i>i</i> is applied
F _{x1i}	Value of the discrete axial load <i>i</i>
N _{qx}	Number of Sections where a distributed axial load is applied
x _{qxi}	X Coordinate where the distributed axial load <i>i</i> is applied

q_{x1i}	Value of the distributed axial load at section i
N_{Pt}	Number of Sections where a discrete transversal load is applied
x_{Pti}	X Coordinate where the discrete transversal load i is applied
P_t	Value of the discrete transversal load i
N_{qt}	Number of Sections where a distributed transversal load is applied
x_{qti}	X Coordinate where the distributed transversal load i is applied
q_{ti}	Value of the distributed transversal load i
N_M	Number of Sections where a discrete bending moment is applied
x_{Mi}	X Coordinate where the discrete bending moment i is applied
M_i	Value of the discrete bending moment i

3 INTRODUCTION

The PANCO Program calculates reserve factors and applied and allowable stresses of uniform and non uniform metallic columns, with different support conditions under the action of axial and/or transversal loads.

A uniform column has a constant cross section along its complete length, and the applied axial load it's only applied at the ends of the beam.

Figure 3-1 below shows the coordinate system.

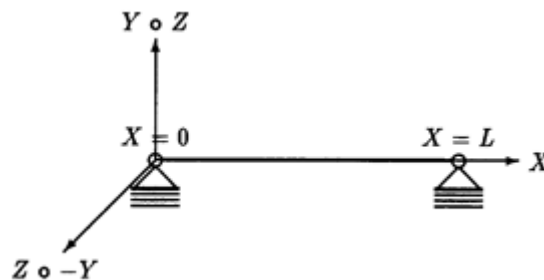


Figure 3-1. Coordinate system

A. The possible support conditions are the following ones:

Uniform Column:

a) Bending:

$X=0$

 Simply supported
 Clamped
 Simply supported
 Free
 Simply supported
 Supported
 Supported

$X=L$

 Simply supported
 Clamped
 Clamped
 Clamped
 Supported
 Clamped
 Supported

b) Torsion:

$X=0$

 Free warping
 Constrained warping
 Free warping
 Constrained warping
 Gyration is always constrained at both ends

$X=L$

 Free warping
 Constrained warping
 Constrained warping
 Free warping

Non Uniform Column:

It is only analysed under bending loads, assuming the same support conditions as the uniform column ones under bending loads.

B. The types of loads that can be applied over the column are the following ones:

Uniform Column:

a) Axial loads:

- Tension or compression load at the ends of the column

b) Transversal loads:

- Discrete loads, applied at any point along the length of the column length
- Distributed loads along the whole length of part of it.
- Discrete bending moments, applied at any point along the column length

Non Uniform Column:

a. Axial loads:

- Discrete loads, applied at any point along the length of the column length
- Distributed loads applied on a part or the whole column length

b. Transversal loads: The same types as for uniform column

C. The types of analysis performed by the program are:

1. Column buckling: The program calculates the Reserve Factor as a column buckling, taking into account an elasto-plastic behaviour of the material. Column buckling loads will be the applied ones multiplied by the reserve factor obtained. Column buckling modes are:

Uniform Column:

- a. Bending buckling respect to Y and Z axis
- b. Torsion buckling
- c. Interaction between bending and crippling
- d. Buckling under bending and torsions loads

Non Uniform Column:

- a. Bending buckling respect to Y axis

2. Beam-Column strength: The program calculates the internal loads, stresses and deflexion of the beam-column resultant of the axial and transversal applied loads, assuming a behaviour completely elastic of the column.

If there are no axial loads applied, the program analyses the beam under transversal loads only.

The program uses the successive approximation method of Newmark to calculate the bending reserve factor and to perform the beam-column analysis. This method follows an iterative process that ends when the relative variations obtained in the last two iterations is lower than 0.5E-04 (result is correct till the fourth significant digit). The compared values are the reserve factor in column buckling and the mean deflexion in the strength analysis.

3.1 DIFFERENCES BETWEEN VERSIONS

- Version 3.x:

Now, the user can choose the language for screen dialogs and for the output file, between Spanish and English.

The data flow of the program is modified. Now, the input data as well as the output results are provided by means of files:

- Input file: With the input data grouped in the same way as for other applications.
- Output file: With all the input data and output results in printable form.

Since the format and arrangement of the input file is completely different from previous version, input files are not compatible between versions.

The new program source code is structured to allow the creation of a Dynamic Link Library DLL for Windows operating system.

In parallel with this, a Windows application has been developed as Graphic User Interface GUI for the program PANCO. This application uses the same input file and provides the same output results. The analysis is performed by calling to the DLL, compiled from the same source as the official Fortran version.

Version 3.3 introduces several bug corrections that improve the program's stability and usability, among others:

- The Windows GUI application can be invoked with the input file as an argument for direct calculation: "`panco.exe example.dat`" creates a file named `example.sal`, located in the same folder as `example.dat`, with results in English.
- The warping constant can be zero.
- Non uniform columns with many sections are calculated correctly.
- Fields in the GUI can be safely modified and are correctly calculated in real time.
- GUI fixed for bending support conditions 5, 6 and 7.
- The RF for Z-axis bending is now correctly calculated and displayed.
- RFs are shown in accordance with Airbus policy (truncated; 2 decimals, "> 2").
- The output file offers now a clearer view of results. For instance, units are displayed.
- Various GUI usability enhancements.

4 PROGRAM EXECUTION

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its full path name.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table); it can be executed by typing the command:

```
$panco
```

The program asks for the language to use, the name of the input file and the name of the output file.

```
Idioma / Language:
=====

1 ... ESPANOL
2 ... ENGLISH
Idioma / Language: 2
Name of the input file: Example.dat
Name of the output file: Results.sal
```

Details of the input and output files are provided in next sections. The CPU time for calculation is negligible.

It is possible to execute the program with a single command, including the input and output filenames and the language as parameters. The above example can be executed as follows:

```
$panco Example.dat Results.sal 2
```

If the third parameter (language) is omitted, the English language will be taken by the program.

5 INPUT DATA FILE

In this chapter, it is described the structure of the input file, which is required for the correct execution of the PANCO program.

This file contains all the necessary data that defines the problem. It basically contains eleven data groups, all of them are necessary. They will be numbered from 1 to 11 and must be in order in the input file.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces. The numbering of each group will be included in the first column.

Note that groups 5, 6 and 7 depend on the value assumed in Group 2.

Listed below are the contents of each data group:

- **GROUP 1: Title of the problem**

Four lines, containing the title of the problem to be solved, which will appear at the header of the output file:

Line 1 1 First line of the title

Line 2 Second line of the title

Line 3 Third line of the title

Line 4 Fourth line of the title

- **GROUP 2: Type of the Column**

Line 1 2 TYPE OF COLUMN: 1 Uniform, 2 Non Uniform

Line 2 1 (if the column is Uniform) or 2 (if the column is Non Uniform)

- **GROUP 3: Units**

Line 1 3 UNITS (N/daN/Kgf/lbf/klb, mm/cm/in)

Line 2 Force Units and Length Units (i.e. N,mm)

- **GROUP 4: Material Data**

Line 1 4 MATERIAL DATA

Line 2 Et Ec G Ftu Fty Fcy e

- **GROUP 5: Geometry**

Line 1 5 GEOMETRY

Then, if Type of Column = 1: Uniform column

Line 2 L A Iyy Izz J R y₀ z₀ Fcc Ds Di

The values have to comply with the following requirements:

$L > 0$

$A > 0$

$I_{yy} \geq 0$ If it is equal to 0, the following analyses will not be performed:

- Bending buckling about Y Axis
- Torsion buckling
- Bending + Torsion buckling
- Buckling about Y axis + Crippling
- Beam-Column Bending about Y axis

$I_{zz} \geq 0$ If it is equal to 0, the following analyses will not be performed:

- Bending buckling about Z Axis
- Torsion buckling
- Bending + Torsion buckling
- Buckling about Z axis + Crippling
- Beam-Column Bending about Z axis

$J \geq 0$ If $J = 0$ and $R = 0$, the following analyses will not be performed:

- Torsion buckling
- Bending + Torsion buckling

$R \geq 0$ If $J = 0$ and $R = 0$, the following analysis will not be performed:

- Torsion buckling
- Bending + Torsion buckling

If $y_0 = 0$, there is no interaction between the buckling about Y axis and Torsion buckling

If $z_0 = 0$, there is no interaction between the buckling about Z axis and Torsion buckling

If $F_{cc} = 0$, the following analysis will not be performed:

- Bending + Crippling buckling

If $D_s = 0$ or $D_i = 0$ the stresses due to the beam-column bending will be zero at the corresponding fiber.

Then, if Type of Column = 2: Non Uniform column

Line 2	N_g	x_1	A_1	I_1	DS_1	DI_1
Line i		x_i	A_i	I_i	DS_i	DI_i
Line N_g		x_{N_g}	A_{N_g}	I_{N_g}	DS_{N_g}	DI_{N_g}

The values have to comply with the following requirements:

$$X \geq 0$$

$$A > 0$$

$I \geq 0$ If $I = 0$ the following analysis will not be performed:

- Bending buckling about Y axis
- Beam-Column bending about Y axis

If $DS_i = 0$ or $DI_i = 0$, the stresses due to the beam-column bending will be zero at the corresponding fiber.

• **GROUP 6:**

Line 1 6 WORK PLANE AND SUPPORT CONDITIONS:

Then, if Type of Column = 1: Uniform column

Line 2 WP (WP = 1 for XZ working plane and WP = 2 for XY working plane)

Line 3 SPy, SPz (support conditions about Y axis Bending and Z axis Bending)

$SP_i = 1$	Simply supported - Simply supported
$SP_i = 2$	Clamped - Clamped
$SP_i = 3$	Simply supported - Clamped
$SP_i = 4$	Free - Clamped
$SP_i = 5$	Simply supported - Supported
$SP_i = 6$	Supported - Clamped
$SP_i = 7$	Supported - Supported

Line 4 SPx (support conditions about X axis Torsion)

$SP_x = 1$	Free warping - Free warping
$SP_x = 2$	Constrained warping - Constrained warping
$SP_x = 3$	Free warping - Constrained warping
$SP_x = 4$	Constrained warping - Free warping

Line 5 Ry1 Ry2 (XZ bending stiffness at support i; line and fields exist if SPy =5, 6 or 7)

Line 6 Rz1 Rz2 (XY bending stiffness at support I; line and fields exist if SPz =5, 6 or 7)

Then, if Type of Column = 2: Non Uniform column

Line 2 SPy (support conditions about Y axis Bending, as above)

Line 3 Ry1 Ry2 (XZ bending stiffness at support i, line and fields exist if SPy =5, 6 or 7)

- **GROUP 7:**

Line 1 7 DISCRETE AXIAL LOADS

Then, if Type of Column = 1: Uniform column

Line 2 1 X_{Px1} F_{x1}

F_{x1} is negative for compression loads and positive for tension loads.

Then, if Type of Column = 2: Non Uniform column

Line 2 N_{Px} X_{Px1} F_{x1}

Line i X_{Pxi} F_{xi}

Line N_{Px} $X_{Px,NPx}$ $F_{x,NPx}$

Positive loads have positive sign of X axis. Data will be typed in ascending order of x coordinate.

- **GROUP 8:**

If Type of Column = 1, then, this Group **must not appear**.

Line 1 8 DISTRIBUTED AXIAL LOADS

Line 2 N_{qx} X_{qx1} q_{x1}

Line i X_{qxi} q_{xi}

Line N_{qx} $X_{qx,Nqx}$ $q_{x,Nqx}$

Positive loads have positive sign of X axis. Data will be typed in ascending order of x coordinate. It will be considered that the load varies linearly along the column length.

Apart from the initial and final sections, the data for Sections in which any change is produced must be written also, and the total length of the column must be represented.

- **GROUP 9:**

Line 1 9 DISCRETE TRANSVERSAL LOADS

Line 2 N_{Pt} X_{Pt1} P_{t1}

Line i X_{Pti} P_{ti}

Line N_{Pt} $X_{Pt,NPt}$ $P_{t,NPt}$

Positive loads have negative sign of Y (or Z) axis. Data will be typed in ascending order of x coordinate.

- **GROUP 10:**

Line 1 10 DISTRIBUTED TRANSVERSAL LOADS

Line 2 N_{qt} X_{qt1} q_{t1}

Line i X_{qti} q_{ti}

Line N_{qt} $X_{qt,Nqt}$ $q_{t,Nqt}$

Positive loads have negative sign of Y (or Z) axis. Data will be typed in ascending order of x coordinate.

Apart from the initial and final sections, the data for Sections in which any change is produced must be written also, and the total length of the column must be represented.

- **GROUP 11:**

Line 1 11 DISCRETE BENDING MOMENTS

Line 2 N_M X_{M1} M_1

Line i X_{Mi} M_i

Line N_M $X_{M,NM}$ M_{NM}

Positive moments have negative (positive) sign of Y (Z) axis for a working plane X. Data will be typed in ascending order of x coordinate.

6 OUTPUT RESULTS

This section includes the description of the output result file as well as the possible error messages.

6.1 Output file

The PANCO program creates one printable file with all the results of the analysis carried out. This file includes the following information:

- **Summary of the input data:**

- Type of column
- Material data
- Geometric Data
- Support conditions
- Applied loads & moments

- **Column Buckling analysis:**

Uniform Column:

- Buckling under Bending about Y axis Reserve Factor
- Buckling under Bending about Z axis Reserve Factor
- Buckling under Torsion about X axis Reserve Factor
- Buckling under Bending + Crippling about Y axis Reserve Factor
- Buckling under Bending + Crippling about Z axis Reserve Factor
- Buckling under Torsion + Bending Reserve Factor

Non Uniform Column:

- Buckling under Bending about Y axis Reserve Factor

For each RF, the program provides as well the corresponding values of the stress and the resultant elastic modulus corrected by plasticity (if it's required).

- **Results:**

Along the length of the column, for the significant X coordinates, the program provides:

- Deflection
- Axial force
- Bending moment
- Shear force
- Stress at Upper fiber
- Stress at Lower fiber

6.2 Error messages

When the program detects an error in the input data, it writes the corresponding message in the console and stops the execution. The error messages that might appear are listed below:

Error messages:

```
ERROR: The input file cannot be opened
ERROR: Erroneous group number in the input file
ERROR: No Group 1 in the input file
ERROR: End of file while reading Group 1 in the input file
ERROR: No Group 2 in the input file
ERROR: End of file while reading Group 2 in the input file
ERROR: Bad definition of Group 2
ERROR: No Group 3 in the input file
ERROR: End of file while reading Group 3 in the input file
ERROR: Bad definition of Group 3
ERROR: No Group 4 in the input file
ERROR: End of file while reading Group 4 in the input file
ERROR: Bad definition of Group 4
ERROR: No Group 5 in the input file
ERROR: End of file while reading Group 5 in the input file
ERROR: Bad definition of Group 5
ERROR: Group 5: Number of Sections higher than 100
ERROR: Group 5: Erroneous column length
ERROR: No Group 6 in the input file
ERROR: End of file while reading Group 6 in the input file
ERROR: Bad definition of Group 6
ERROR: Group 6: Bad definition of working plane
ERROR: Group 6: Support in Y axis bad defined
ERROR: Group 6: Support in Z axis bad defined
ERROR: Group 6: Support in Torsion bad defined
ERROR: End of file while reading group 7 in the input file
ERROR: Bad definition of Group 7
ERROR: Group 7: Number of Sections higher than 100
ERROR: Group 7: Erroneous number of Sections
ERROR: End of file while reading group 8 in the input file
ERROR: Bad definition of Group 8
ERROR: Group 8: Number of Sections higher than 100
ERROR: Group 8: Erroneous number of Sections
ERROR: End of file while reading group 9 in the input file
ERROR: Bad definition of Group 9
ERROR: Group 9: Number of Sections higher than 100
ERROR: Group 9: Erroneous number of Sections
ERROR: End of file while reading group 10 in the input file
ERROR: Bad definition of Group 10
ERROR: Group 10: Number of Sections higher than 100
ERROR: Group 10: Erroneous number of Sections
ERROR: End of file while reading group 11 in the input file
ERROR: Bad definition of Group 11
ERROR: Group 11: Number of Sections higher than 100
ERROR: Group 11: Erroneous number of Sections
ERROR: No applied loads
ERROR: Undefined working plane
ERROR: Undefined support conditions
```

ERROR: Et = 0
ERROR: Ec = 0
ERROR: G = 0
ERROR: Ftu = 0
ERROR: Fty = 0
ERROR: Fcy = 0
ERROR: e = 0
ERROR: Undefined type of analysis
ERROR: Erroneous geometric data
ERROR: Iyy = 0 o Izz = 0
ERROR: Maximum deflection is null
ERROR: Area calculation: Low Limit > Up Lim
ERROR: Area calculation: Disordered abscissas
ERROR: Area calculation: Low Limit out of interval
ERROR: Area calculation: Up Limit out of interval
ERROR: Area calculation: Different sub-intervals
ERROR: Convergence failure obtaining the root
ERROR: Null Maximum stress
ERROR: The output file cannot be opened
ERROR: The output file cannot be written
ERROR: The abscissas are not sorted
ERROR: Abscissa to be interpolated is out of interval

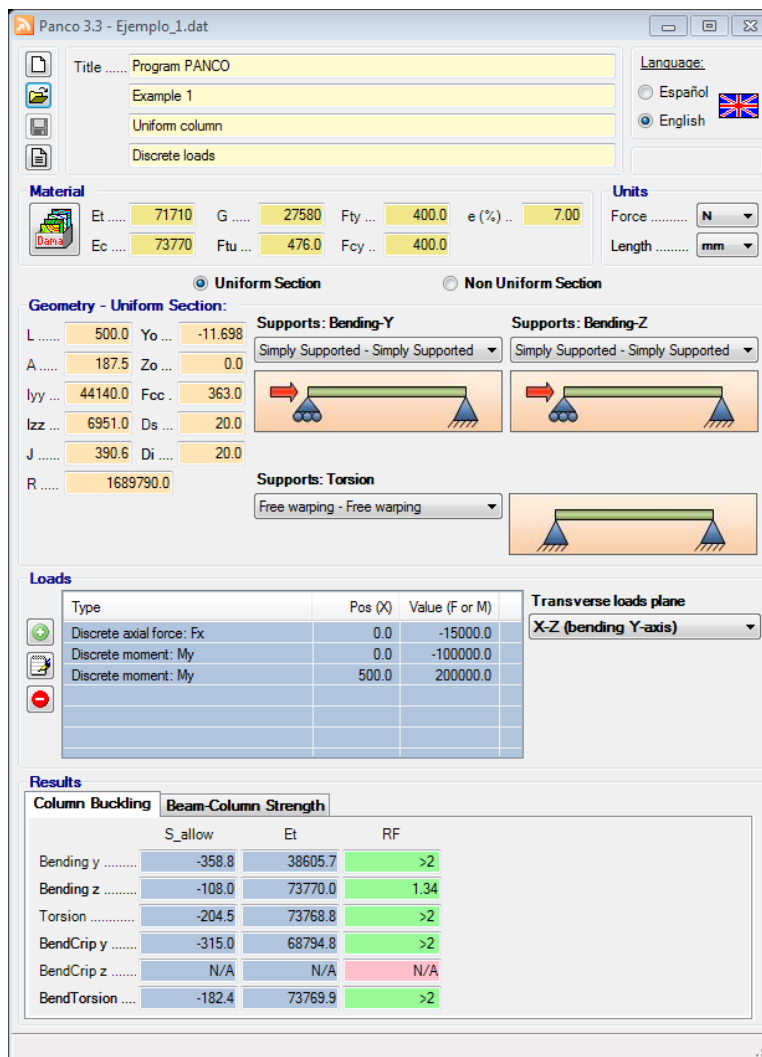
7 WINDOWS APPLICATION

An application has been developed in Visual Basic .NET for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please contact with the system administrator.

The application can be executed a) "double clicking" in the file Panco.exe located in the PC's server, b) using the Calculo menu located in the Windows taskbar, or c) by means of a link in the "desktop".

The input data must be typed in the application form or retrieved from an input file. The form, after loading the data corresponding to the example 1 (see chapter 8), has the following appearance:



The screenshot shows the 'Panco 3.3 - Ejemplo_1.dat' application window. It contains several sections for input data:

- Title:** Program PANCO, Example 1, Uniform column, Discrete loads.
- Language:** English (selected).
- Material:** Et: 71710, G: 27580, Fty: 400.0, e (%): 7.00, Ec: 73770, Ftu: 476.0, Fcy: 400.0.
- Units:** Force: N, Length: mm.
- Geometry - Uniform Section:** L: 500.0, Yo: -11.698, A: 187.5, Zo: 0.0, Iyy: 44140.0, Fcc: 363.0, Izz: 6951.0, Ds: 20.0, J: 390.6, Di: 20.0, R: 1689790.0.
- Supports:** Bending-Y: Simply Supported - Simply Supported, Bending-Z: Simply Supported - Simply Supported, Torsion: Free warping - Free warping.
- Loads:**

Type	Pos (X)	Value (F or M)
Discrete axial force: Fx	0.0	-15000.0
Discrete moment: My	0.0	-100000.0
Discrete moment: My	500.0	200000.0
- Results:**

	S _{allow}	Et	RF
Bending y	-358.8	38605.7	>2
Bending z	-108.0	73770.0	1.34
Torsion	-204.5	73768.8	>2
BendCrip y	-315.0	68794.8	>2
BendCrip z	N/A	N/A	N/A
BendTorsion	-182.4	73769.9	>2

Figure 7-1. Input data form

The program calculates the results for all the included loads at the moment it has all the required data. In example 1, there are one axial load and a couple of discrete bending moments and the user can read the results at the bottom framebox.

8 EXAMPLES

Input data and obtained results are presented for the two following examples.

8.1 Example 1

In the first example it will be analysed a Uniform Column, simply supported at both ends. The applied loads are showed in Figure 3 below:

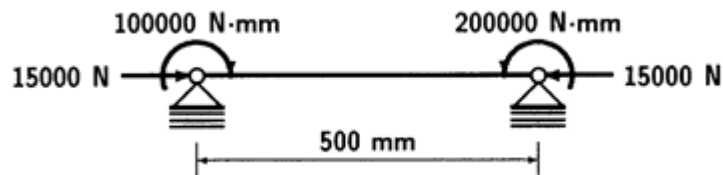


Figure 8-1. Example 1

The material properties of the column are:

$E_t =$	71710 N/mm ²
$E_c =$	73770 N/mm ²
$G =$	27580 N/mm ²
$F_{tu} =$	476 N/mm ²
$F_{ty} =$	400 N/mm ²
$F_{cy} =$	400 N/mm ²
$e =$	7 %

The length of the column is $L = 500$ mm, and the cross section geometric properties are:

$A =$	187.5 mm ⁴
$I_{yy} =$	44140.0 mm ⁴
$I_{zz} =$	6951.0 mm ⁴
$J =$	390.6 mm ⁴
$R =$	1689790.0 mm ⁴
$Y_o =$	-11.698 mm
$Z_o =$	0.0 mm
$F_{cc} =$	363.0 N/mm ²
$D_s =$	20.0 mm
$D_i =$	20.0 mm

Input file:

```
1 Program PANCO
  Example 1
  Uniform column
  Discrete loads
2 TYPE OF COLUMN: 1 Uniform, 2 Non Uniform
  1
3 UNITS: N/daN/Kgf/lbf/klb   mm/cm/in
  N   , mm
4 MATERIAL DATA: Et Ec G Ftu Fty Fcy Delta
  71710, 73770, 27580, 476.0, 400.0, 400.0, 7.00
5 GEOMETRY: [ Uniform => Xlong Area Iyy Izz Jtors Cwarp Ycc Zcc Scrip
  DisNaUF DisNaLF ]   [ Non Uniform => Nx X Area Iyy DisNaUF
  DisNaLF ]
  500.0, 187.5, 44140, 6951, 390.6, 1689790, -11.698, 0.0, 363.0, 20.0, 20.0
6 WORK PLANE AND SUPPORT CONDITIONS: [ Uniform => Plane (1-XZ/2-XY) / Supports
  (Bend-Y, Bend-Z / Tors-X) / Stiffness (Y, Z) ]   [ Non Uniform => Bend-Y /
  Stiff-Y ]
  1
  1, 1
  1
7 DISCRETE AXIAL LOADS
  1,      0.00,      -15000.0
11 DISCRETE BENDING MOMENTS
  2,      0.00,      -100000.00
      500.00,      200000.00
```

Output file:

```
*****
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*   <<< PROGRAM PANCO >>>         *
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Buckling and strength analysis for columns under axial and/or lateral load.

TITLE:
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Program PANCO
Example 1
Uniform column
Discrete loads

INPUT DATA:
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Type of column:

Uniform (constant section and discrete axial load applied at both ends)

Material properties of the column:

Tension elastic modulus:	Et =	71710.00 N/mm ²
Compression elastic modulus:	Ec =	73770.00 N/mm ²
Shear elastic modulus:	G =	27580.00 N/mm ²
Ultimate tension stress:	Ftu =	476.00 N/mm ²
Tension yield stress:	Fty =	400.00 N/mm ²
Compression yield stress:	Fcy =	400.00 N/mm ²
Permanent strain at ultimate stress:	e =	7.00 %

Geometrical data of the column:

Column length:	L =	500.00 mm
Cross sectional area:	A =	187.50 mm ²
Moment of inertia about Y-axis:	Iyy =	44140.00 mm ⁴
Moment of inertia about Z-axis:	Izz =	6951.00 mm ⁴
Torsion constant:	J =	390.60 mm ⁴
Warping constant:	R =	1689790.00 mm ⁶
Y coordinate from Shear Center to CG:	Yo =	-11.70 mm
Z coordinate from Shear Center to CG:	Zo =	0.00 mm
Crippling failure stress of the column:	Fcc =	363.00 N/mm ²
Distance from neutral axis to upper fibre:	Ds =	20.00 mm
Distance from neutral axis to lower fibre:	Di =	20.00 mm

Support conditions at both ends of the column:

	Xs = 0	Xs = L
Flex. rel. to Y-axis:	Simply supported	Simply supported
Flex. rel. to Z-axis:	Simply supported	Simply supported
Torsion:	: Free warping	Free warping

Turn at both ends is not allowed in all cases

Applied loads on the column:

XS (mm)	AXIAL LOADS		TRANSVERSAL LOADS		
	Px (N)	px (N/mm)	Pt (N)	pt (N/mm)	Mf (N·mm)
0.000	15000.00	0.00	0.00	0.00	-100000.00
500.000	-15000.00	0.00	0.00	0.00	200000.00

COLUMN BUCKLING ANALYSIS:

Stresses, tangent moduli and reserve factors for buckling loads:

BUCKLING MODE	S (N/mm ²)	Et (N/mm ²)	R.F.
Bending about Y - axis	-358.8	38605.7	>2
Bending about Z - axis	-108.0	73770.0	1.34
Torsion	-204.5	73768.8	>2
Bending-Y + Crippling	-315.0	68794.8	>2
Bending-Z + Crippling	---	---	---
Bending-Y + Torsion	-182.4	73769.9	>2

STRENGTH ANALYSIS AS BEAM-COLUMN:

Maximum and minimum summary:

		MAXIMUM		MINIMUM	
		XS (mm)	VALUE	XS (mm)	VALUE
Deflection (DELTA)		0.000	0.00 mm	262.500	-1.67 mm
Axial force (FA)		0.000	-15000.00 N	0.000	-15000.00 N
Bending moment (MF)		487.500	200106.44 N·mm	0.000	100000.00 N·mm
Shear force (FC)		0.000	374.73 N	500.000	-14.36 N
Up fibre stress (S_FS)		0.000	-125.31 N/mm ²	487.500	-170.67 N/mm ²
Lw fibre stress (S_FI)		487.500	10.67 N/mm ²	0.000	-34.69 N/mm ²

Deflection, forces, bending moments and stresses corresponding to the applied loads:

XS (mm)	DELTA (mm)	FA (N)	MF (N·mm)	FC (N)	S_FS (N/mm ²)	S_FI (N/mm ²)
0.000	0.00000	-15000.00	100000.00	374.73	-125.31	-34.69
12.500	-0.14307	-15000.00	104645.97	368.69	-127.42	-32.58
25.000	-0.28104	-15000.00	109215.55	362.51	-129.49	-30.51
37.500	-0.41370	-15000.00	113705.41	356.00	-131.52	-28.48
50.000	-0.54083	-15000.00	118112.27	349.23	-133.52	-26.48
62.500	-0.66221	-15000.00	122432.93	342.20	-135.47	-24.53
75.000	-0.77764	-15000.00	126664.22	334.93	-137.39	-22.61
87.500	-0.88689	-15000.00	130803.05	327.41	-139.27	-20.73
100.000	-0.98979	-15000.00	134846.42	319.65	-141.10	-18.90
112.500	-1.08612	-15000.00	138791.36	311.66	-142.89	-17.11
125.000	-1.17570	-15000.00	142634.98	303.44	-144.63	-15.37
137.500	-1.25834	-15000.00	146374.52	295.00	-146.32	-13.68
150.000	-1.33385	-15000.00	150007.20	286.34	-147.97	-12.03
162.500	-1.40206	-15000.00	153530.41	277.47	-149.57	-10.43
175.000	-1.46281	-15000.00	156941.53	268.41	-151.11	-8.89
187.500	-1.51592	-15000.00	160238.12	259.14	-152.60	-7.40
200.000	-1.56122	-15000.00	163417.75	249.69	-154.05	-5.95
212.500	-1.59858	-15000.00	166478.09	240.06	-155.43	-4.57
225.000	-1.62784	-15000.00	169416.94	230.24	-156.76	-3.24
237.500	-1.64885	-15000.00	172232.11	220.27	-158.04	-1.96
250.000	-1.66148	-15000.00	174921.58	210.13	-159.26	-0.74
262.500	-1.66560	-15000.00	177483.38	199.84	-160.42	0.42
275.000	-1.66108	-15000.00	179915.62	189.40	-161.52	1.52
287.500	-1.64781	-15000.00	182216.55	178.82	-162.56	2.56
300.000	-1.62567	-15000.00	184384.48	168.11	-163.55	3.55
312.500	-1.59455	-15000.00	186417.83	157.28	-164.47	4.47
325.000	-1.55437	-15000.00	188315.09	146.34	-165.33	5.33
337.500	-1.50502	-15000.00	190074.91	135.29	-166.12	6.12
350.000	-1.44643	-15000.00	191696.00	124.14	-166.86	6.86
362.500	-1.37850	-15000.00	193177.16	112.89	-167.53	7.53
375.000	-1.30117	-15000.00	194517.30	101.57	-168.14	8.14
387.500	-1.21438	-15000.00	195715.45	90.17	-168.68	8.68

400.000	-1.11807	-15000.00	196770.75	78.71	-169.16	9.16
412.500	-1.01218	-15000.00	197682.45	67.19	-169.57	9.57
425.000	-0.89667	-15000.00	198449.83	55.62	-169.92	9.92
437.500	-0.77150	-15000.00	199072.36	44.01	-170.20	10.20
450.000	-0.63665	-15000.00	199549.58	32.36	-170.42	10.42
462.500	-0.49208	-15000.00	199881.16	20.70	-170.57	10.57
475.000	-0.33779	-15000.00	200066.81	9.01	-170.65	10.65
487.500	-0.17377	-15000.00	200106.44	-2.67	-170.67	10.67
500.000	0.00000	-15000.00	200000.00	-14.36	-170.62	10.62

8.2 Example 2

In the second Example, it will analyzed a Non Uniform column, fixed at both ends. The applied loads are showed in Figure 3 below:

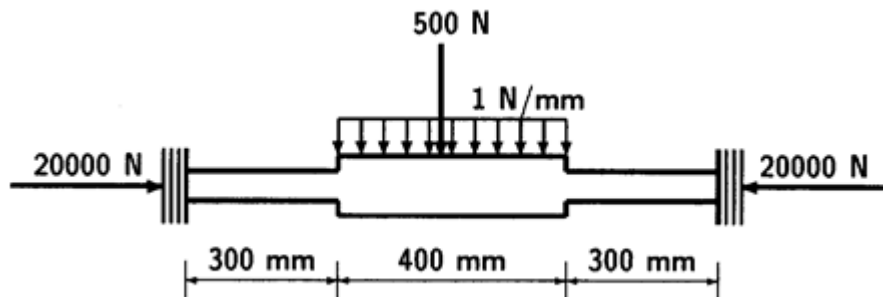


Figure 8-2. Example 2

The material properties of the column are:

$E_t =$	71710 N/mm
$E_c =$	73770 N/mm ²
$G =$	27580 N/mm ²
$F_{tu} =$	476 N/mm ²
$F_{ty} =$	400 N/mm ²
$F_{cy} =$	400 N/mm ²
$e =$	7 %

The length of the column is $L = 1000$ mm, and the cross section geometric properties are:

- Normal section:

$A =$	250 mm ²
$I_{yy} =$	15000 mm ⁴
$D_s =$	25 mm
$D_i =$	25 mm

- Reinforced section:

$A =$	400 mm ²
$I_{yy} =$	25000 mm ⁴
$D_s =$	40 mm
$D_i =$	40 mm

Input file:

```

1 Program PANCO
  Example 2
  Not uniform column
  Discrete and distributed loads
2 TYPE OF COLUMN: 1 Uniform, 2 Non Uniform
  2
3 UNITS: N/daN/Kgf/lbf/klb   mm/cm/in
  N, mm
4 MATERIAL DATA: Et Ec G Ftu Fty Fcy Delta
  71710, 73770, 25580, 476.0, 400.0, 400.0, 7.00
5 GEOMETRY: [ Uniform => Xlong Area Iyy Izz Jtors Cwarp Ycc Zcc Scrip
  DisNaUF DisNaLF ]   [ Non Uniform => Nx X Area Iyy DisNaUF
  DisNaLF ]
  6,      0.00,      250.0,      15000.0,      25.00,      25.00
      300.00,      250.0,      15000.0,      25.00,      25.00
      300.00,      400.0,      25000.0,      40.00,      40.00
      700.00,      400.0,      25000.0,      40.00,      40.00
      700.00,      250.0,      15000.0,      25.00,      25.00
      1000.00,      250.0,      15000.0,      25.00,      25.00
6 WORK PLANE AND SUPPORT CONDITIONS: [ Uniform => Plane (1-XZ/2-XY) / Supports
  (Bend-Y, Bend-Z / Tors-X) / Stiffeness (Y, Z) ]   [ Non Uniform => Bend-Y /
  Stiff-Y ]
  2
7 DISCRETE AXIAL LOADS
  2,      0.00,      20000.0
      1000.00,      -20000.0
9 DISCRETE TRANSVERSAL LOADS
  1,      500.00,      500.0
10 DISTRIBUTED TRANSVERSAL LOADS
  6,      0.00,      0.0
      300.00,      0.0
      300.00,      1.0
      700.00,      1.0
      700.00,      0.0
      1000.00,      0.0

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Output file:

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*                                     *
*   <<< PROGRAM PANCO >>>         *
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Buckling and strength analysis for columns under axial and/or lateral load.

TITLE:
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Program PANCO
Example 2
Not uniform column
Discrete and distributed loads

INPUT DATA:
=====

Type of column:

Not uniform (variable section and discrete axial load applied on several sections)

Material properties of the column:

Tension elastic modulus:	Et =	71710.00 N/mm ²
Compression elastic modulus:	Ec =	73770.00 N/mm ²
Shear elastic modulus:	G =	25580.00 N/mm ²
Ultimate tension stress:	Ftu =	476.00 N/mm ²
Tension yield stress:	Fty =	400.00 N/mm ²
Compression yield stress:	Fcy =	400.00 N/mm ²
Permanent strain at ultimate stress:	e =	7.00 %

Geometrical data of the column:

XS (mm)	AREA (mm ²)	INERTIA (mm ⁴)	DISTANCE FROM NEUTRAL AXIS TO	
			UPPER FIBRE (mm)	LOWER FIBRE (mm)
0.00000	250.00	15000.00	25.000	25.000
299.99960	250.00	15000.00	25.000	25.000
300.00000	400.00	25000.00	40.000	40.000
699.99957	400.00	25000.00	40.000	40.000
700.00000	250.00	15000.00	25.000	25.000
1000.00000	250.00	15000.00	25.000	25.000

Support conditions at both ends of the column:

	Xs = 0	Xs = L
	-----	-----
Flex. rel. to Y-axis:	Clamped	Clamped
	Turn at both ends is not allowed in all cases	

Applied loads on the column:

XS (mm)	AXIAL LOADS		TRANSVERSAL LOADS		
	Px (N)	px (N/mm)	Pt (N)	pt (N/mm)	Mf (N · mm)
0.000	20000.00	0.00	0.00	0.00	0.00
300.000	0.00	0.00	0.00	0.00	0.00
300.000	0.00	0.00	0.00	1.00	0.00
500.000	0.00	0.00	500.00	1.00	0.00
700.000	0.00	0.00	0.00	1.00	0.00

700.000	0.00	0.00	0.00	0.00	0.00
1000.000	-20000.00	0.00	0.00	0.00	0.00

COLUMN BUCKLING ANALYSIS:

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BUCKLING MODE	R.F.
Bending about Y-axis	>2

Stresses and tangent moduli corresponding to buckling loads:

XS (mm)	S (N/mm ²)	Et (N/mm ²)
0.000	-226.05	73761.5
25.000	-226.05	73761.5
50.000	-226.05	73761.5
75.000	-226.05	73761.5
100.000	-226.05	73761.5
125.000	-226.05	73761.5
150.000	-226.05	73761.5
175.000	-226.05	73761.5
200.000	-226.05	73761.5
225.000	-226.05	73761.5
250.000	-226.05	73761.5
275.000	-226.05	73761.5
300.000	-226.05	73761.5
300.000	-141.28	73770.0
325.000	-141.28	73770.0
350.000	-141.28	73770.0
375.000	-141.28	73770.0
400.000	-141.28	73770.0
425.000	-141.28	73770.0
450.000	-141.28	73770.0
475.000	-141.28	73770.0
500.000	-141.28	73770.0
525.000	-141.28	73770.0
550.000	-141.28	73770.0
575.000	-141.28	73770.0
600.000	-141.28	73770.0
625.000	-141.28	73770.0
650.000	-141.28	73770.0
675.000	-141.28	73770.0
700.000	-141.28	73770.0
700.000	-226.05	73761.5
725.000	-226.05	73761.5
750.000	-226.05	73761.5
775.000	-226.05	73761.5
800.000	-226.05	73761.5
825.000	-226.05	73761.5
850.000	-226.05	73761.5
875.000	-226.05	73761.5
900.000	-226.05	73761.5
924.999	-226.05	73761.5
949.998	-226.05	73761.5
974.997	-226.05	73761.5
1000.000	-226.05	73761.5

STRENGTH ANALYSIS AS BEAM-COLUMN:

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Maximum and minimum summary:

		MAXIMUM		MINIMUM	
		XS (mm)	VALUE	XS (mm)	VALUE
Deflection	(DELTA)	999.999	0.00 mm	500.000	-4.87 mm
Axial force	(FA)	0.000	-20000.00 N	0.000	-20000.00 N

Bending moment (MF)	500.000	160420.12 N·mm	0.000	-141972.84 N·mm
Shear force (FC)	225.000	754.83 N	775.000	-754.83 N
Up fibre stress (S_FS)	0.000	156.62 N/mm ²	500.000	-306.67 N/mm ²
Lw fibre stress (S_FI)	500.000	206.67 N/mm ²	0.000	-316.62 N/mm ²

Deflection, forces, bending moments and stresses corresponding to the applied loads:

XS (mm)	DELTA (mm)	FA (N)	MF (N·mm)	FC (N)	S_FS (N/mm ²)	S_FI (N/mm ²)
0.000	0.00000	-20000.00	-141972.84	451.84	156.62	-316.62
25.000	-0.03951	-20000.00	-129932.62	513.04	136.55	-296.55
50.000	-0.15345	-20000.00	-116403.86	571.07	114.01	-274.01
75.000	-0.33398	-20000.00	-101543.45	621.21	89.24	-249.24
100.000	-0.57260	-20000.00	-85521.12	664.25	62.54	-222.54
125.000	-0.86015	-20000.00	-68520.26	699.74	34.20	-194.20
150.000	-1.18695	-20000.00	-50734.42	727.21	4.56	-164.56
175.000	-1.54278	-20000.00	-32368.16	746.32	-26.05	-133.95
200.000	-1.91711	-20000.00	-13631.75	754.83	-57.28	-102.72
225.000	-2.29925	-20000.00	5260.79	754.83	-88.77	-71.23
250.000	-2.67838	-20000.00	24093.05	752.08	-120.16	-39.84
275.000	-3.04372	-20000.00	42649.76	735.22	-151.08	-8.92
300.000	-3.38465	-20000.00	60718.03	712.96	-181.20	21.20
300.000	-3.38465	-20000.00	60718.30	711.78	-147.15	47.15
325.000	-3.69849	-20000.00	77932.36	666.39	-174.69	74.69
350.000	-3.98554	-20000.00	93985.74	618.86	-200.38	100.38
375.000	-4.24035	-20000.00	108769.23	565.84	-224.03	124.03
400.000	-4.45781	-20000.00	122180.78	508.95	-245.49	145.49
425.000	-4.63334	-20000.00	134128.77	448.52	-264.61	164.61
450.000	-4.76277	-20000.00	144529.83	385.02	-281.25	181.25
475.000	-4.84260	-20000.00	153313.88	318.43	-295.30	195.30
500.000	-4.86979	-20000.00	160420.03	250.69	-306.67	206.67
500.000	-4.86979	-20000.00	160420.12	-250.65	-306.67	206.67
525.000	-4.84263	-20000.00	153314.42	-318.43	-295.30	195.30
550.000	-4.76277	-20000.00	144529.89	-385.06	-281.25	181.25
575.000	-4.63332	-20000.00	134128.30	-448.52	-264.61	164.61
600.000	-4.45781	-20000.00	122180.75	-508.91	-245.49	145.49
625.000	-4.24037	-20000.00	108769.63	-565.84	-224.03	124.03
650.000	-3.98555	-20000.00	93985.91	-618.88	-200.38	100.38
675.000	-3.69847	-20000.00	77932.17	-666.40	-174.69	74.69
700.000	-3.38466	-20000.00	60718.56	-711.75	-147.15	47.15
700.000	-3.38465	-20000.00	60718.24	-712.96	-181.20	21.20
725.000	-3.04373	-20000.00	42650.02	-735.22	-151.08	-8.92
750.000	-2.67838	-20000.00	24093.20	-752.09	-120.16	-39.84
775.000	-2.29924	-20000.00	5260.72	-754.83	-88.77	-71.23
800.000	-1.91711	-20000.00	-13631.70	-754.83	-57.28	-102.72
825.000	-1.54276	-20000.00	-32368.46	-746.33	-26.05	-133.95
850.000	-1.18695	-20000.00	-50734.39	-727.19	4.56	-164.56
875.000	-0.86016	-20000.00	-68519.94	-699.74	34.20	-194.20
900.000	-0.57260	-20000.00	-85521.00	-664.29	62.54	-222.54
924.999	-0.33395	-20000.00	-101543.43	-621.22	89.24	-249.24
949.998	-0.15346	-20000.00	-116402.60	-571.02	114.00	-274.00
974.997	-0.03955	-20000.00	-129930.31	-513.03	136.55	-296.55
1000.000	0.00000	-20000.00	-141972.70	-451.90	156.62	-316.62